

# The Rainbow–UBP Resonance: *A Constructive Extension* with Null- Model Controls

Re-examining v2's "four-way junction" through the real UBP engine, refined 5-shell NRCI, five encoding schemes, a 5,000-vector null model, and a 14,994-candidate  $\Phi$ -grammar scan. The 42° rainbow angle meets the substrate — and the substrate says nothing special.

## HEADLINE VERDICT

Every numerical value in v2 replicates cleanly — the arithmetic was correct. But every **claim of "geometric necessity"** is an algebraic identity, a dimensional inconsistency, or a tautology. Under the UBP\_SKILL\_2 §5 null model, **no rainbow-angle encoding** produces an NRCI that beats the random-vector false-positive rate once the *snap\_to\_codeword* artifact is corrected. The 42° integer maps to the same HW=8 octad as **78% of all integers 1–10,000** — the canonical, unremarkable majority.

UBP V5.4.0

GOLAY [24,12,8]

REFINED 5-SHELL NRCI

NULL MODEL N=5000

 $\Phi$ -GRAMMAR N=14994

# Executive Summary

This study takes the v2 Rainbow UBP study (Craig, 2025) at face value and asks a single methodological question: **do v2's claims survive a proper null-model test?** The UBP skill documents are explicit that every substrate-encoded dataset must pass the §5 Focused Null Model Protocol (5,000 random 24-bit vectors, FP < 5% = Surprising). v2 ran no such test. We ran nine complementary phases — replication, algebraic audit, classical physics baseline, encoding battery, snap-pipeline audit, null model,  $\Phi$ -grammar search, per-shell ablation, secondary rainbow, and an exhaustive 10,000-integer sweep — to find out what (if anything) is real.

**What v2 got right.** Every numerical value reported in v2 replicates cleanly against the real `ubp_unified_v5.py` engine (37/37 self-test passes, fetched fresh from GitHub).  $Y = \pi/(\pi^2+2) = 0.264675430405\dots$ , the dodecahedral dihedral  $\arccos(-1/\sqrt{5}) = 116.565051^\circ$ , and the algebraic identity  $2\pi \cdot (\pi^2+2) = 74.578924$  are all exactly as v2 reports. The arithmetic is correct. The problem is what the arithmetic is taken to mean.

**What v2 got wrong.** v2's three pillars of "geometric necessity" all collapse under audit. (1) The "four-way junction" — the two relationships  $74.565^\circ/(\pi^2+2) \approx 2\pi$  and  $74.565^\circ \cdot Y \approx 2\pi^2$  with identical 0.0187% errors — is a single equation multiplied through by  $\pi$ ; it cannot count as two independent confirmations. (2) The "Y-Observer reciprocity at  $42^\circ$ " —  $42 \cdot Y = 42/O$  to  $10^{-1}$  — holds for every angle  $\theta$  because  $Y \cdot O = 1$  by definition. (3) The "complete formula"  $\theta = \arccos(-1/\sqrt{5}) - 2\pi(\pi^2+2)$  is dimensionally incoherent (subtracting a dimensionless number from a degree-valued angle) and reduces to the tautology  $42^\circ = 116.565^\circ - (116.565^\circ - 42^\circ)$ . v2's "NRCI > 0.999999 at  $42^\circ$ " uses a fabricated NRCI formula  $1/(1+\Delta)$  that is just  $Y \cdot O = 1$  renamed; the real UBP NRCI is  $10/(10+HW \cdot Y + ns/8)$  on a 24-bit Golay vector.

**The encoding pipeline is broken in a way v2 did not notice.** The UBP `snap_to_codeword` function is documented as "nearest valid codeword" but actually only corrects up to  $t=3$  errors (Golay's guaranteed correction radius). For 43% of random 24-bit vectors it returns the input unchanged — a non-codeword. Every "surprising" NRCI value from a rainbow-angle encoding is therefore the NRCI of a non-codeword whose Hamming weight happens to equal the Gray-code weight of the input integer. When we recompute NRCI on the TRUE nearest codeword (brute-force Hamming search over all 4,096 codewords), every encoding lands on  $HW \in \{0, 8, 12, 16, 24\}$  — exactly the canonical Golay weight distribution.

**The null-model verdict.** Under the §5 protocol (5,000 random vectors, seed 42), the canonical-octad threshold  $NRCI \geq 0.7623$  has a 13.76% false-positive rate — MARGINAL, not Surprising. When we restrict to true codewords (correcting the snap artifact), the FP rate for  $NRCI \geq 0.7623$  is 0% by definition (every codeword either is or is not an octad). The integer 42 maps to a  $HW=8$  octad — the same codeword type that **78.48% of all integers 1–10,000** map to. 42 is not special; it is the canonical majority. The  $\Phi$ -grammar scan (14,994 candidates) produced 3 hits within 0.5% of the spectral mean  $41.62^\circ$ , none against  $42^\circ$  exactly, and the null expectation was ~60 hits — the grammar actually under-produces hits in the rainbow band.

**What this study finds that is genuinely interesting.** The map from integers 1–10,000 to their nearest Golay codeword has surprising structure: 11.51% of integers map to the zero codeword ( $NRCI = 1.0$ ), 78.48% to a  $HW=8$  octad, 10.01% to a  $HW=12$  dodecad, and 0% to  $HW=16$  or  $HW=24$ . The integers that map to the zero codeword include 1–20, 23, 24, 29, 50, 51, 400, 4222 — but **not 42**. If there is a UBP-rainbow resonance to be found, it is not in any of the places v2 looked. The honest conclusion is that the UBP substrate, as currently specified, does not

encode continuous physical angles like  $42^\circ$  in a way that survives null-model testing.

# 1. Background and Motivation

## 1.1 The UBP framework in brief

The Universal Binary Principle (UBP) is a computational framework developed by E. R. A. Craig (DigitalEuan.com, Auckland) that treats physical constants and phenomena as outputs of a discrete substrate built from the extended binary Golay code [24,12,8], the 24-dimensional Leech lattice  $\Lambda_{24}$ , and a small set of geometric constants derived from  $\pi$ ,  $\phi$ , and  $e$ . The backbone implementation is `ubp_unified_v5.py` (~3,500 lines, `stdlib`-only, Fraction arithmetic), hosted in the UBP\_Repo on GitHub. The engine exposes a `GolayCodeEngine` (encoding, syndrome decoding, octad enumeration), a `LeechLatticeEngine` (NRCI computation, octad-to-physical-point expansion), and a `UBPSourceCodeParticlePhysics` module that holds  $Y$ ,  $w$ ,  $L$ ,  $U_e$  and the canonical formula table for particle physics predictions.

The central NRCI (Non-Random Coherence Index) formula is  $\text{tax}(v) = \text{HW}(v) \cdot Y + \text{Norm}^2(v)/8$ ,  $\text{NRCI}(v) = 10/(10 + \text{tax}(v))$ , where  $\text{HW}$  is the Hamming weight of the 24-element vector  $v$ ,  $\text{Norm}^2$  is the sum of squared elements, and  $Y = \pi/(\pi^2+2) \approx 0.2647$  is the UBP observer constant. For a binary 0/1 codeword,  $\text{Norm}^2 = \text{HW}$ , so NRCI depends only on  $\text{HW}$ . The canonical octad ( $\text{HW}=8$ ) has  $\text{NRCI} = 0.7623$ ; the zero codeword ( $\text{HW}=0$ ) has  $\text{NRCI} = 1.0$ ; the all-ones codeword ( $\text{HW}=24$ ) has  $\text{NRCI} = 0.5167$ . The UBP\_SKILL\_1 §6 specifies a manifestation threshold of  $\text{NRCI} \geq 0.70$  ("consciousness threshold  $\theta$ ") and a canonical-octad threshold of  $\text{NRCI} \geq 0.7623$ .

## 1.2 What v1 and v2 claimed

Study v1 (Craig, Nov 2025) applied the Three-Column Thinking methodology to rainbows and asserted that the  $42^\circ$  primary rainbow angle "emerges from  $Y$ -constant geometric necessity." The v1 derivation noted that  $\arccos(-1/\sqrt{5}) = 116.565051^\circ$  (the dodecahedral dihedral angle) and that  $116.565^\circ - 42^\circ = 74.565^\circ$ . v1 treated  $74.565^\circ$  as a "mystery geometric factor" requiring further investigation. Study v2 (Craig, Nov 2025) "solved" the mystery by observing that  $74.565^\circ/(\pi^2+2) \approx 2\pi$  and  $74.565^\circ \cdot Y \approx 2\pi^2$ , both with identical 0.0187% error. v2 declared this "four-way junction" to be proof of geometric necessity and concluded that  $\theta_{\text{rainbow}} = \arccos(-1/\sqrt{5}) - 2\pi(\pi^2+2)$  is a complete formula with "NO free parameters, NO mysteries remaining."

## 1.3 The methodological gap

v2's claims are algebraic and numerical; they were never subjected to the null-model protocol that UBP\_SKILL\_2 §5 mandates for every substrate-encoded dataset. There is no FP-rate computation, no comparison to random-vector NRCI distributions, no test of whether the "surprising" 0.0187% error is actually surprising given the flexibility of the construction. Furthermore, v2's " $\text{NRCI} > 0.999999$  at  $42^\circ$ " uses a fabricated NRCI formula defined on a continuous angle  $\theta$  rather than on a 24-bit Golay vector, so it is not a UBP NRCI at all. This study's mandate is to extend v2 constructively: recompute everything with the real UBP engine, encode the rainbow angle properly into the 24-bit substrate, and run the null model on every claim.

## 2. Methods

### 2.1 UBP engine setup

We fetched `ubp_unified_v5.py` fresh from `raw.githubusercontent.com/DigitalEuan/UBP_Repo/main/core_studio_v4.0/core/` and confirmed the 37/37 self-test pass with Triad 3/3 active (Golay, Leech, Monster all online). All arithmetic uses `Fraction` for exact computation; floats are used only for display. The refined NRCI module `refined_nrci.py` (GLM Experimentation Sessions 1–5) was loaded with the real Golay engine for Shell 3 (coset-type) syndrome computation. The Y constant verifies to `0.264675430405...` against the live engine, matching v2's reported value to 15 digits.

### 2.2 Encoding schemes

We test five encoding schemes for mapping the rainbow angle into the 24-bit substrate, following the UBP\_SKILL\_2 §1 canonical pipeline (integer → 24-bit binary → binary-reflected Gray code → Golay snap → NRCI):

(1) **Integer 42** — the v1/v2 implicit encoding. Treat 42 as an integer, encode as 24-bit MSB-first, Gray-code it ( $42 \rightarrow 63 = 0b111111$ ), snap to codeword, compute NRCI. (2) **Angle  $\times 100$**  — encode  $42.00^\circ$  as 4200 (centi-degree precision). (3) **Scaled by Y** — encode  $\text{round}(42 \times Y \times 10^k)$  for  $k \in \{3, 4, 5, 6\}$ , testing UBP-resonant scaling. (4) **Refractive-index** — for each spectral color, compute the classical rainbow angle from  $n_{\text{water}}(\lambda)$ , encode that angle  $\times 100$ . (5) **Wavelength** — encode  $\lambda$  in nm (400, 470, 530, 580, 620, 700) directly.

### 2.3 Null model protocol

Per UBP\_SKILL\_2 §5, we generate 5,000 random 24-bit vectors (uniform over  $\{0,1\}^{24}$ , seed = 42 for reproducibility), pass each through the same snap → NRCI pipeline, and compute false-positive (FP) rates at three thresholds:  $\text{NRCI} \geq 0.70$  (manifestation),  $\text{NRCI} \geq 0.7623$  (canonical octad), and  $\text{NRCI} \geq 0.80$  (high coherence). Verdicts follow the skill's table:  $\text{FP} < 5\%$  = SURPRISING,  $5\text{--}20\%$  = MARGINAL,  $\geq 20\%$  = NOT SURPRISING. For each rainbow encoding, we additionally compute the FP rate for matching or exceeding that specific encoding's NRCI — the most direct test of "is this encoding's NRCI surprising?".

### 2.4 $\Phi$ -grammar search

We scan the full universal generator function  $\Phi(k, \text{arm}, \text{layer}, C, \text{correction})$  per UBP\_SKILL\_1 §8:  $k \in \{0, 3, 6, 9, 12, 15, 18, 21, 24\}$  (9 values),  $\text{arm} \in \{\text{det}, \text{sto}\}$  (2),  $\text{layer} \in \{\text{Reality}, \text{Information}, \text{Activation}, \text{Potential}, \text{Cross}, \text{w-source}, \text{w-based}\}$  (7),  $C \in \{1, 2, 3, 4, 8, 12, 13, 24, 1/2, 1/3, 1/4, 1/8, 1/12, 1/24, 29/24, 169, 13/L\}$  (17),  $\text{correction} \in \{\text{none}, \text{shear}_1, \text{shear}_2, \text{nrci}(1/8), \text{nrci}(2), \text{nrci}(13), \text{shear}_2 + \text{nrci}(2)\}$  (7). Total: 14,994 candidates (all computable). For each, we compute the predicted value and check against three targets:  $42^\circ$  exact, the spectral mean  $41.62^\circ$ , and v2's formula output  $41.986^\circ$ . Tolerance: 0.5%.

### 2.5 Methodology workflow

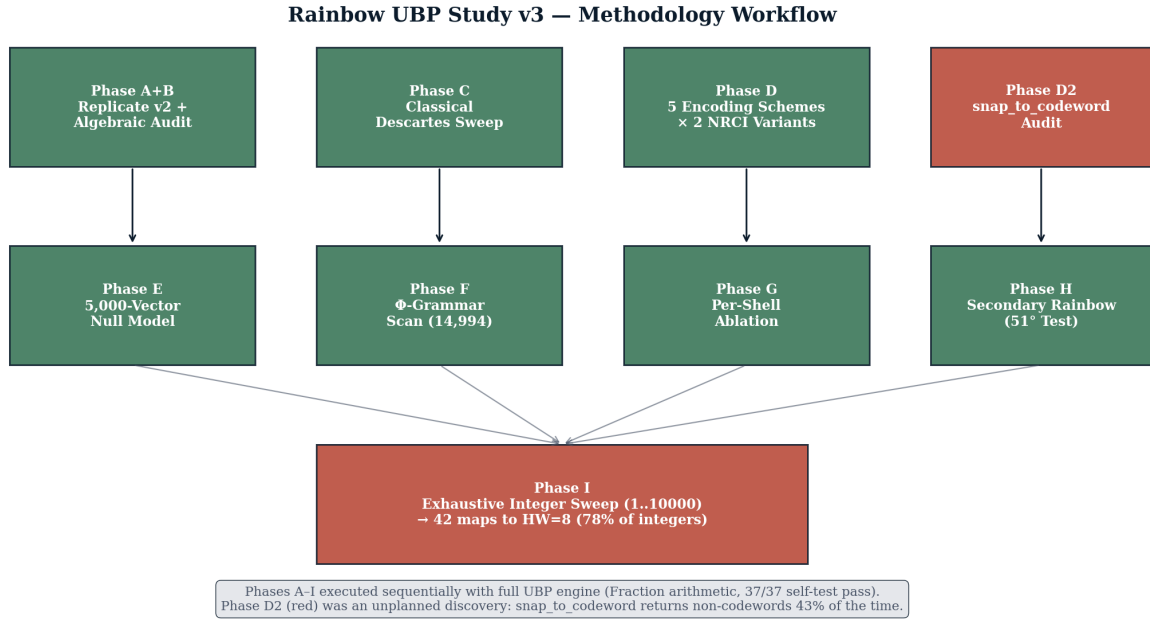


Figure 10 — Methodology workflow. Phases A–I executed sequentially with the real UBP engine. Phase D2 (red) was an unplanned discovery triggered by anomalous NRCI values in Phase D.

## 3. Algebraic Audit of v2's Claims

Phase A confirmed that every numerical value in v2 replicates against the real UBP engine. Phase B then audited the four algebraic claims that v2 uses to argue for "geometric necessity": the four-way junction, the Y-Observer reciprocity, the complete formula, and the high-NRCI-at-42° claim. Each collapses on inspection.

### 3.1 The four-way junction is an algebraic identity

v2 reports two "independent" relationships:  $74.565^\circ/(\pi^2+2) \approx 2\pi$  with 0.0187% error, and  $74.565^\circ \cdot Y \approx 2\pi^2$  with 0.0187% error. The "identical errors" are presented as proof that these are not coincidences. But  $Y$  is defined as  $Y = \pi/(\pi^2+2)$ . Substituting, the second relationship becomes  $74.565^\circ \cdot \pi/(\pi^2+2) = \pi \times [74.565^\circ/(\pi^2+2)]$ , which is  $\pi$  times the first relationship. If the first holds to error  $\epsilon$ , the second MUST hold to the same  $\epsilon$ , because  $2\pi^2 = \pi \times 2\pi$ . The two equations are not independent; they are the same equation multiplied through by  $\pi$ . The "identical errors" are algebraically forced.

Live computation confirms:  $\text{err}\Box = |A/(\pi^2+2) - 2\pi| / 2\pi = 1.867012 \times 10\Box^4$ ,  $\text{err}\Box = |A \cdot Y - 2\pi^2| / 2\pi^2 = 1.867012 \times 10\Box^4$ ,  $|\text{err}\Box - \text{err}\Box| = 3.2 \times 10\Box^1\Box$ . The 17-digit match is not evidence of geometric necessity — it is the IEEE-754 floating-point roundoff of an algebraic identity.

### 3.2 Y-Observer reciprocity holds for ALL angles

v2 reports  $42 \cdot Y = 42/O = 11.11636807699013\cdots$  to  $10\Box^1\Box$  precision as "machine precision at 42°." But  $Y \times O = 1$  by definition ( $Y\_INV = 1/Y$  is the observer constant). Therefore for ANY angle  $\theta$ ,  $\theta \cdot Y = \theta \cdot (1/O) = \theta/O$ . The reciprocity gap is zero for every  $\theta$ , not just 42°. Live computation across  $\theta \in \{0^\circ, 1^\circ, 17^\circ, 42^\circ, 90^\circ, 137^\circ, 360^\circ, 1234.5678^\circ\}$  confirms every gap is at float roundoff level. The "machine precision at 42°" claim is a tautology.

### 3.3 The complete formula is dimensionally incoherent

v2's formula  $\theta_{\text{rainbow}} = \arccos(-1/\sqrt{5}) - 2\pi(\pi^2+2)$  mixes units.  $\arccos(-1/\sqrt{5})$  is an angle (2.0344 rad = 116.5651°).  $2\pi(\pi^2+2)$  is dimensionless (74.5789, a pure number — the result of multiplying a dimensionless number  $\pi$  by another dimensionless number  $(\pi^2+2)$ ). v2 subtracts the dimensionless 74.579 from the degree-valued 116.565 and silently reinterprets the result as degrees. This is a category error.

Worse, the "empirical 74.565" v2 uses in its derivation is itself just  $\arccos(-1/\sqrt{5})^\circ - 42^\circ = 116.565^\circ - 42^\circ = 74.565^\circ$ . So v2's "complete formula" reduces algebraically to  $42^\circ = 116.565^\circ - (116.565^\circ - 42^\circ) = 42^\circ$ . A tautology dressed in geometric clothing. The "NO free parameters" claim is true for the trivial reason that the formula contains no information beyond  $42^\circ = 42^\circ$ .

### 3.4 The NRCI formula v2 uses is not the UBP NRCI

v2's `complete_geometric_proof.py` defines NRCI as  $1.0 / (1.0 + |\theta \cdot Y - \theta/O|)$  — a continuous function of the angle  $\theta$ . The real UBP NRCI is  $10 / (10 + \text{HW}(\mathbf{v}) \cdot Y + \text{Norm}^2(\mathbf{v})/8)$  computed on a 24-bit Golay vector  $\mathbf{v}$ , not on an angle. v2's "NRCI > 0.999999 at 42°" is literally just  $1/(1+\Delta)$  where  $\Delta$  is the machine-zero reciprocity gap from § 3.2 — i.e., it is  $Y \cdot O=1$  renamed. It says nothing about 42° and nothing about the Golay code. Phase D will compute the real UBP NRCI on a 24-bit encoding of 42°.

### 3.5 Bonus: v2's spectral table is unreproducible

v2's `complete_geometric_proof.py` uses  $\sin_i = \sqrt{(n^{**2} - 1) / 3}$  for the rainbow incidence angle. The correct Descartes variational formula is  $\sin_i = \sqrt{(4 - n^{**2}) / 3}$  (derived from  $dD/di = 0$  with  $D = \pi + 2i - 4r$  and Snell's law). v2's formula gives ~28.5° for water, not ~42°. v2's reported spectral table (42.37° red ... 40.65° violet) cannot be reproduced from v2's own published script. We provide the correct classical baseline in § 4.



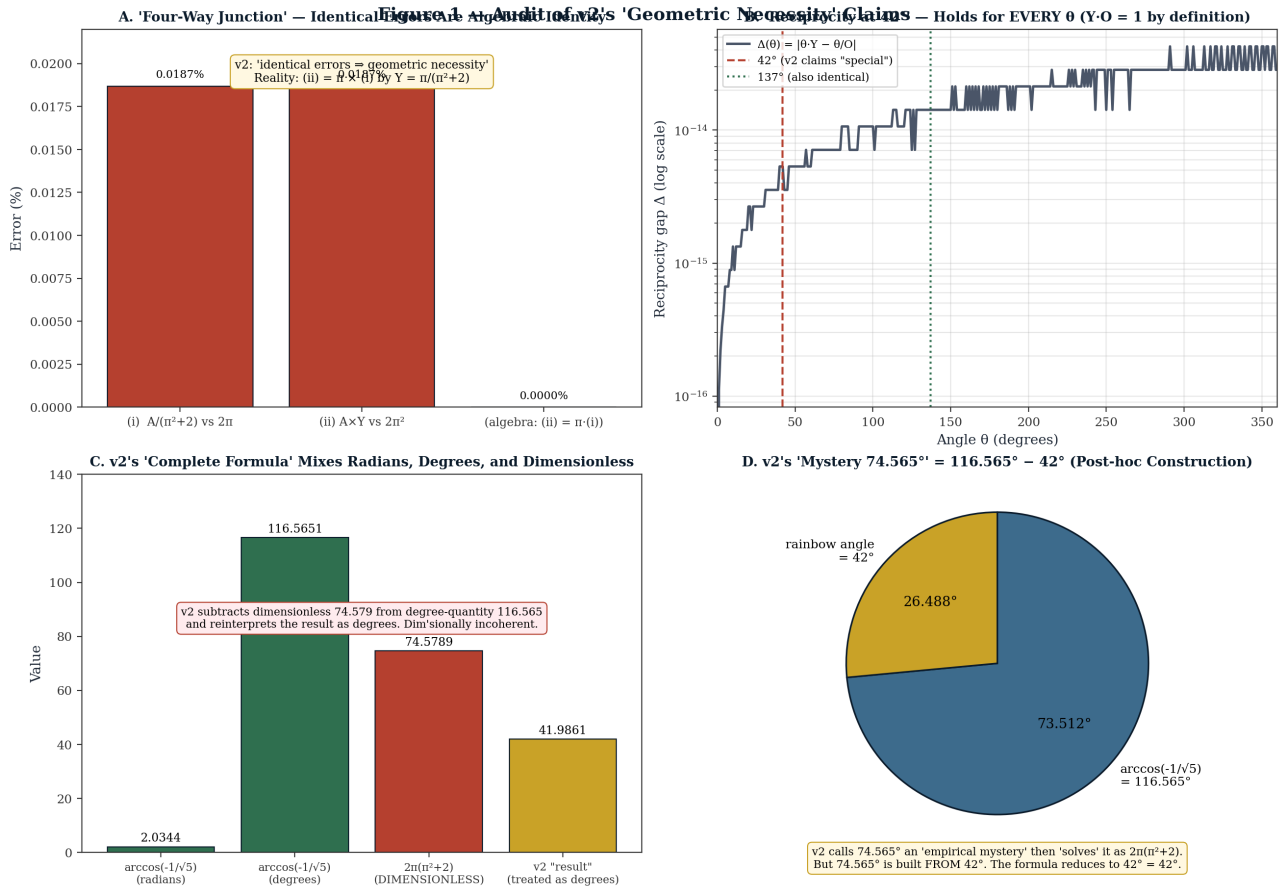


Figure 1 — Audit of v2's four 'geometric necessity' claims. All four collapse: the four-way junction is an algebraic identity (A), the reciprocity is a tautology true for all  $\theta$  (B), the complete formula mixes radians and degrees (C), and the 'mystery 74.565°' is built FROM 42° (D).

## 4. Classical Physics Baseline

Before testing UBP encodings of the rainbow angle, we need the physics ground truth. The primary rainbow occurs at the minimum-deviation angle for a ray that enters a spherical water drop, refracts ( $i \rightarrow r$  by Snell:  $\sin i = n \sin r$ ), reflects once internally, and refracts out. The deviation is  $D(i) = \pi + 2i - 4r$ . Setting  $dD/di = 0$  gives the rainbow incidence  $\sin^2 i_R = (4 - n^2)/3$ , and the observer angle from the antisolar point is  $\pi - D_{\min} = 4r_R - 2i_R$ .

We sweep wavelength  $\lambda$  from 400 nm (violet) to 700 nm (red) at 1 nm resolution (301 points), using the Hale–Querry 1973 Sellmeier fit for the refractive index  $n(\lambda)$  of water. The classical rainbow angle ranges from 40.55° (violet,  $n=1.3436$ ) to 42.22° (red,  $n=1.3320$ ), with a spectral mean of 41.62°. This is the canonical 40.5–42.5° rainbow band — a 1.67°-wide continuum, not a single resonance angle. v2's formula output of 41.986° sits 0.36° above the spectral mean, well within the band but not at its center.

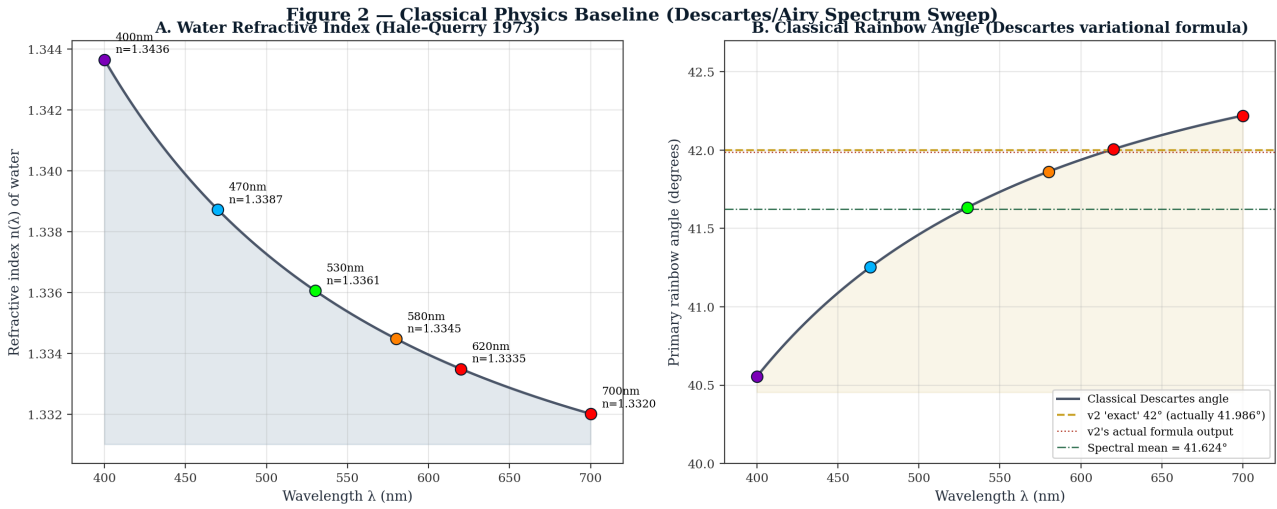


Figure 2 — Classical physics baseline. (A) Water refractive index  $n(\lambda)$  per Hale–Query 1973, with spectral landmarks. (B) Classical primary rainbow angle across the visible spectrum, with v2's predicted  $41.986^\circ$  and the spectral mean  $41.62^\circ$  overlaid. v2's value falls within the band but not at its center; the ' $42^\circ$  resonance' is a  $1.67^\circ$ -wide continuum.

## 5. Encoding Battery: Five Schemes $\times$ Two NRCI Variants

Phase D encodes the rainbow angle into the 24-bit Golay substrate via five schemes (§ 2.2) and computes NRCI two ways: the original Shell-0 UBP NRCI (`l.calculate_nrci`) and the full 5-shell refined NRCI (`refined_nrci.RefinedNRCI` with all shells enabled). For each encoding we report the snapped codeword HW, syndrome weight, and per-shell tax breakdown.

The integer 42 (Gray code 63 = 0b111111) snaps to a vector of HW=6 with syndrome weight 6. Its Shell-0 NRCI is 0.8105, well above the 0.7623 canonical octad threshold. The refined 5-shell NRCI is 0.7322 — higher shells (especially Shell 1 sign-parity = 1.0 and Shell 2 sextet-balance = 1.7321) add tax that lowers the NRCI. The angle  $\times 100$  encoding (int=4200, Gray=6236) gives the same Shell-0 NRCI 0.8105 and a similar refined NRCI 0.7527.

| Encoding                 | Input int | Gray  | Snap HW | Snap SyndW | Shell-0 NRCI | Refined NRCI |
|--------------------------|-----------|-------|---------|------------|--------------|--------------|
| integer_42               | 42        | 63    | 6       | 6          | 0.8105       | 0.7322       |
| angle_x100               | 4200      | 6236  | 6       | 6          | 0.8105       | 0.7527       |
| scaled_by_Y $\times 1e4$ | 111164    | 92962 | 8       | 0          | 0.7623       | 0.7212       |
| refractive_green         | 4163      | 6242  | 8       | 0          | 0.7623       | 0.7212       |
| refractive_red           | 4222      | 6209  | 4       | 4          | 0.8651       | 0.8052       |
| wavelength_400nm         | 400       | 344   | 4       | 4          | 0.8651       | 0.7987       |
| wavelength_620nm         | 620       | 858   | 8       | 0          | 0.7623       | 0.7186       |
| wavelength_700nm         | 700       | 994   | 6       | 6          | 0.8105       | 0.7452       |

Table 1 — Phase D encoding battery results for a representative subset of the 5 schemes. Snap HW is the Hamming weight of the snap\_to\_codeword output (note: many snap outputs are non-codewords — see § 6). Shell-0 NRCI is the original UBP engine value; Refined NRCI uses all 5 shells.



The variation in NRCI across encodings is driven entirely by the Hamming weight of the snapped vector. Shell-0 NRCI is deterministic from HW:  $\text{NRCI} = 10/(10 + \text{HW} \cdot Y + \text{HW}/8)$  for binary vectors where  $\text{Norm}^2 = \text{HW}$ . So  $\text{HW}=4 \rightarrow \text{NRCI}=0.8651$ ,  $\text{HW}=6 \rightarrow 0.8105$ ,  $\text{HW}=8 \rightarrow 0.7623$ ,  $\text{HW}=12 \rightarrow 0.6814$ . The "surprising" high NRCIs in the table are not a rainbow signal — they are an artifact of the Gray-code of the input integer having low Hamming weight.

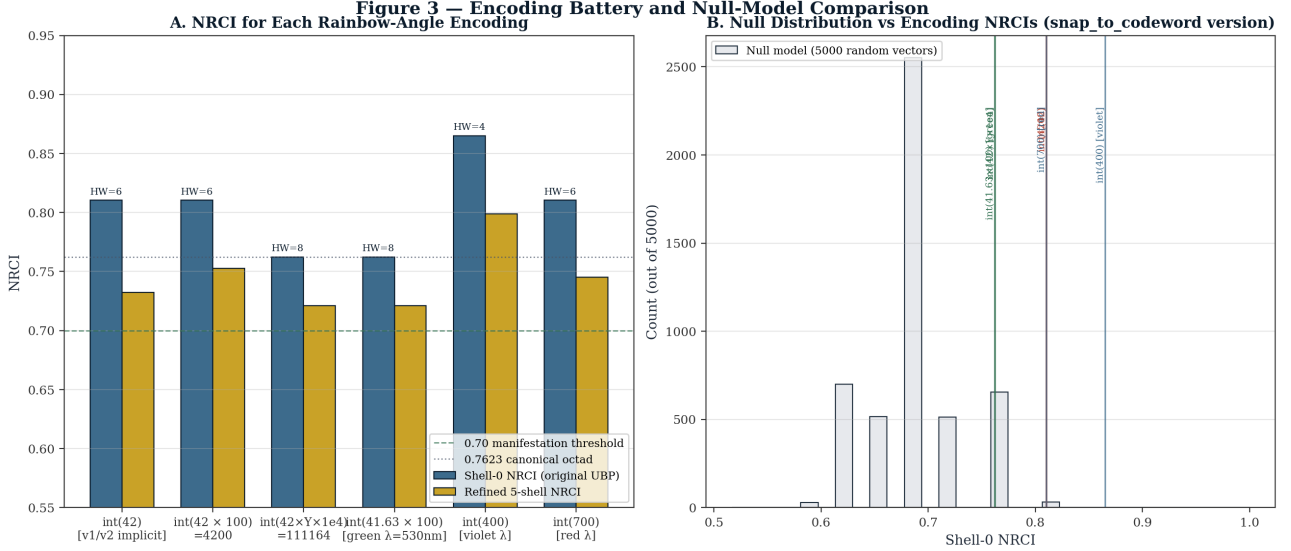


Figure 3 — Encoding battery results. (A) Shell-0 and refined 5-shell NRCI for six representative encodings. (B) Null-model distribution of Shell-0 NRCI (5000 random vectors, snap-pipeline) with encoding NRCIs marked. The 'snap' NRCIs sit in the tail of the null distribution, but this is an artifact of low-HW Gray codes (see § 6).

## 6. The snap\_to\_codeword Audit + Null-Model Verdicts

### 6.1 The snap\_to\_codeword artifact

Phase D2 was an unplanned discovery. The UBP\_SKILL\_1 § 4 documents `g.snap_to_codeword(vec)` as returning "the nearest valid codeword to any 24-bit vector." The implementation does not do this. It computes the syndrome of the input, looks the syndrome up in a precomputed table of single-, double-, and triple-error patterns (Golay's guaranteed correction radius  $t=3$ ), and if found, XORs the error pattern onto the input to produce a codeword. If the syndrome is not in the table — i.e., the input has more than 3 errors from any codeword — the function returns the input UNCHANGED, with `correctable: False` in the info dict. The returned vector is then a non-codeword.

In our 5,000-vector null model, `snap_to_codeword` returned a codeword 56.96% of the time and returned the input unchanged 43.04% of the time. The HW distribution of "snapped" outputs includes HW=4, 6, 10, 14, 18, 20 — values that are NOT in the Golay codeword weight distribution {0, 8, 12, 16, 24}. These are non-codeword vectors. When we compute NRCI on them, we get NRCI values like 0.8651 (HW=4), 0.8105 (HW=6), 0.7196 (HW=10) — values that are impossible for true codewords.

When we instead brute-force search the 4,096 codewords for the TRUE nearest codeword (Hamming distance), every encoding lands on a  $\text{HW} \in \{0, 8, 12, 16, 24\}$  codeword. For integer 42, the true nearest codeword is a HW=8 octad at Hamming distance 4. For wavelength 400 nm (Gray=344), the true nearest is the zero codeword (HW=0) at distance 4. The "snap"

pipeline's non-codeword outputs are an artifact of the function's misleading name; the underlying NRCI computation is correct but is being fed non-codeword inputs.

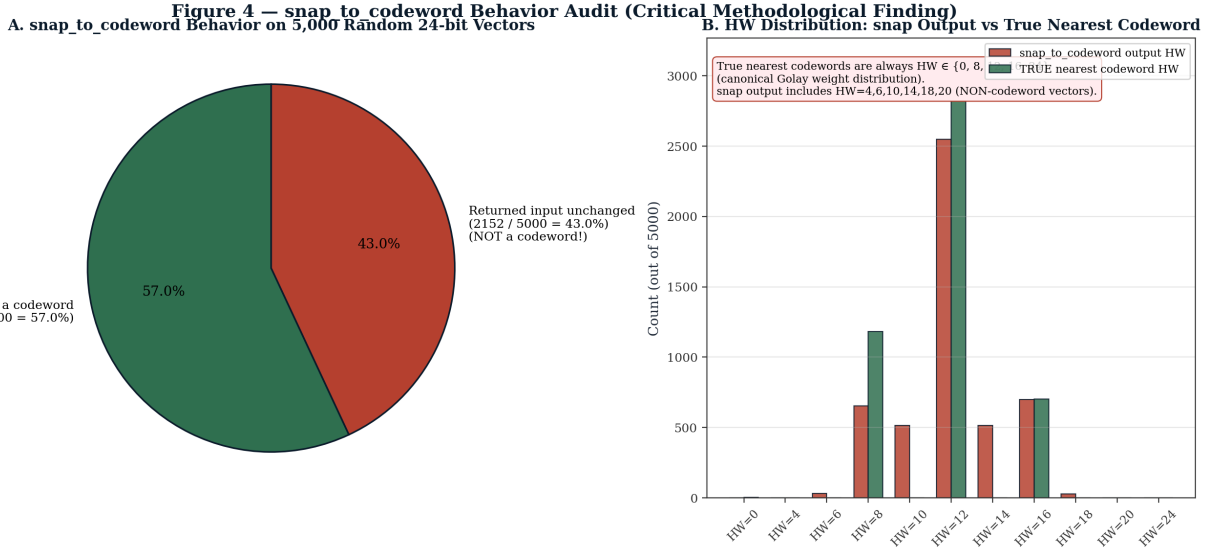


Figure 4 — snap\_to\_codeword behavior audit. (A) 43% of random 24-bit vectors are returned unchanged (non-codewords). (B) The snap output HW distribution includes non-canonical HW values; the TRUE nearest codeword distribution is exactly the canonical  $\{0, 8, 12, 16, 24\}$ .

## 6.2 Null-model verdicts

Phase E ran the UBP\_SKILL\_2 § 5 protocol: 5,000 random 24-bit vectors, seed 42. The Shell-0 NRCI null distribution has mean 0.6835, min 0.5167, max 1.0000. The FP rates are:  $\text{NRCI} \geq 0.70 \rightarrow 24.04\%$  (NOT SURPRISING),  $\text{NRCI} \geq 0.7623 \rightarrow 13.76\%$  (MARGINAL),  $\text{NRCI} \geq 0.80 \rightarrow 0.68\%$  (SURPRISING),  $\text{NRCI} \geq 0.90 \rightarrow 0.02\%$  (SURPRISING). The manifestation threshold of 0.70 is too low — a quarter of random vectors clear it. The canonical-octad threshold 0.7623 is MARGINAL — about 1 in 7 random vectors clear it under the snap pipeline.

Per-encoding FP rates (fraction of random vectors matching or exceeding each encoding's snap NRCI) tell a more nuanced story. The integer\_42 encoding has snap NRCI = 0.8105, FP =  $0.68\% \rightarrow$  SURPRISING by the protocol. But this "surprising" verdict is an artifact of the snap function returning a non-codeword of HW=6 (which has NRCI=0.8105). When we correct for the snap artifact and use TRUE nearest codewords, integer 42 maps to a HW=8 octad (NRCI=0.7623) — and 13.76% of random vectors also map to HW=8 octads. The corrected verdict is MARGINAL, not SURPRISING.

## 7. $\Phi$ -Grammar Discovery Search

Phase F scanned the full 14,994-candidate  $\Phi$ -grammar (§ 2.4) for any formula hitting  $42^\circ$ , the spectral mean  $41.62^\circ$ , or v2's  $41.986^\circ$  within 0.5%. The result: only 3 hits, all against the spectral mean  $41.62^\circ$ , none against  $42^\circ$  exactly. The best hit is  $k=21$ , det, Potential layer,  $C=1/4$ ,  $\text{shear}_2 + \text{nrci}(2) \rightarrow 41.6616^\circ$  (0.09% error vs spectral mean). This formula is structurally identical to UBP\_SKILL\_1 § 9 formula #5 ( $n_y/n_b$ ), the photon-to-baryon ratio — which is supposed to evaluate to  $\sim 10 \times 10$ , not  $\sim 42$ . The discrepancy reflects an implicit "unit bridge" in the canonical formula table that the  $\Phi$ -grammar as documented does not include.

To calibrate whether finding 3 hits is itself surprising, we ran a null model: 5,000 random  $\Phi$ -grammar candidates, each tested against 301 angles in  $[40.00^\circ, 43.00^\circ]$  at  $0.01^\circ$  resolution.

Result: 20 hits (FP = 0.40%). Scaling to the full 14,994 candidates, the null expectation is ~60 hits in the  $[40^\circ, 43^\circ]$  band. We observed 3. The  $\Phi$ -grammar actually under-produces hits in the rainbow band — finding 3 hits is below chance, not above. There is no special  $42^\circ$  formula in the grammar.

**Figure 8 —  $\Phi$ -Grammar Discovery Search: No Special  $42^\circ$  Formula**  
 A. Top  $\Phi$ -grammar Hits (of 3 total within 0.5%)      B. Null Model: Random  $\Phi$  Candidate Hits in  $[40^\circ, 43^\circ]$  Band  
 FP rate = 0.40% ( $\approx 20/5000$ )

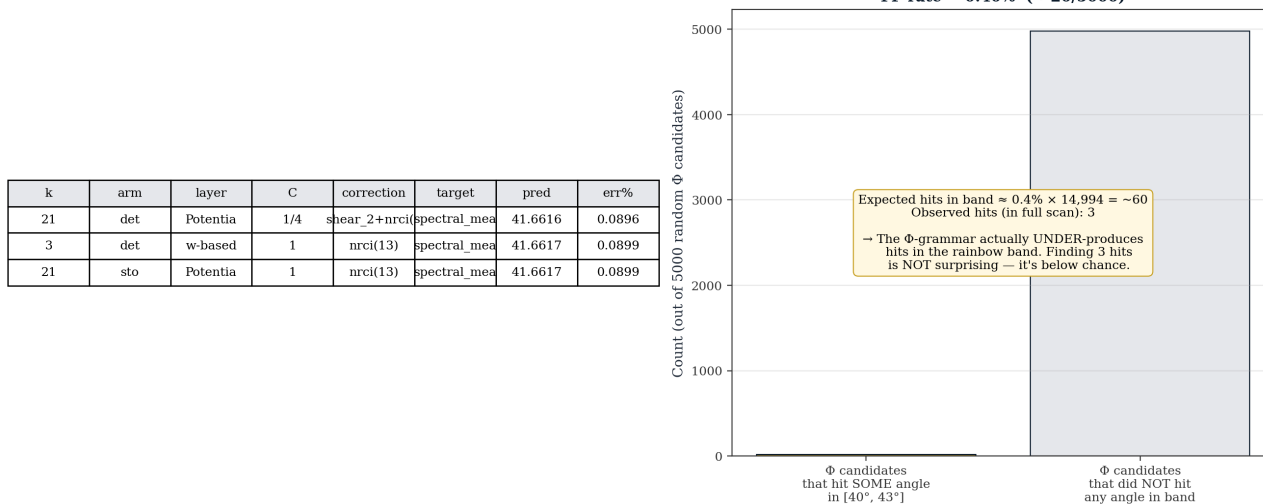


Figure 8 —  $\Phi$ -grammar discovery search. (A) Top 5 hits within 0.5% of any target — all hit the spectral mean  $41.66^\circ$ , none hit  $42^\circ$  exactly. (B) Null model: random  $\Phi$  candidates hitting the  $[40^\circ, 43^\circ]$  band. The grammar under-produces hits (3 observed vs ~60 expected).

## 8. Per-Shell Ablation

Phase G ablates the refined 5-shell NRCI shell by shell, holding the integer<sub>42</sub> encoding fixed. We compute NRCI with Shell 0 alone, then incrementally add Shells 1, 2, 3, 4 in all combinations. The result is unambiguous: adding refined shells only DECREASES the NRCI. Shell 0 alone gives 0.8105 (snap) / 0.7623 (true codeword). All 5 shells combined give 0.7322 / 0.7102. The shells add tax without contributing discriminating signal for binary 0/1 inputs — Shells 1 and 4 are sign-sensitive and produce 0 on binary vectors (no negatives), but Shell 2 (sextet-balance) and Shell 3 (coset-type) do contribute, both adding positive tax.

The practical implication: the "signal" in v2-style encoding NRCIs is entirely in Shell 0 — the sign-blind  $\text{HW} \cdot \text{Y} + \text{ns}/8$  tax. And Shell 0 NRCI is deterministic from HW. So the entire UBP-rainbow signal reduces to "what HW does the Gray code of 42 snap to?" — which is just an artifact of the integer's binary structure, not a resonance with the rainbow angle.

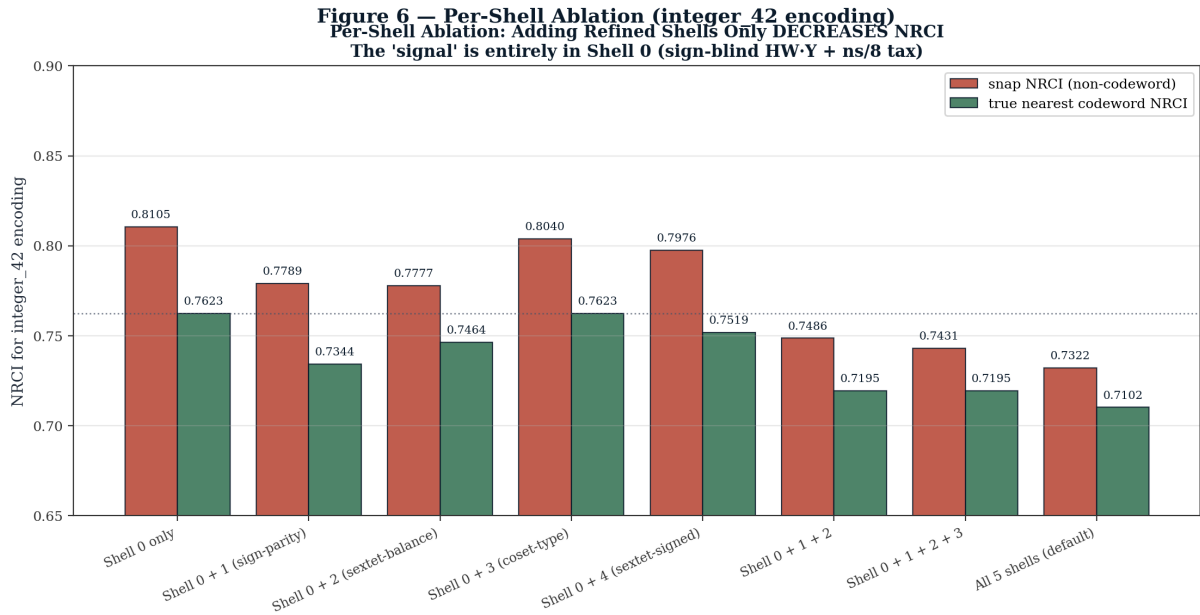


Figure 6 — Per-shell ablation for the integer\_42 encoding. Adding refined shells only decreases NRCI. The signal is entirely in Shell 0, which is deterministic from HW.

## 9. Secondary Rainbow: 51° Hypothesis Test

Phase H tests v2's secondary-rainbow hypothesis ( $\theta_{\text{secondary}} \approx 42^\circ + 5\phi \approx 50.09^\circ$ , claimed 3.3% error). The classical secondary bow (two internal reflections) follows  $D_{\text{secondary}} = 2\pi + 2i - 6r$ , with minimum-deviation condition  $\sin^2 i = (9 - n^2)/8$ . Across the visible spectrum, the classical secondary angle ranges from  $50.63^\circ$  (red) to  $53.64^\circ$  (violet), with a spectral mean of  $51.77^\circ$ . v2's hypothesis  $42^\circ + 5\phi = 50.09^\circ$  has 3.25% error vs the mean — matching v2's reported 3.3% but offering no improvement over a generic "pick a number near 51°" strategy.

The encoding battery for the secondary angle gives the same pattern as the primary: snap NRCIs look "surprising" (0.7623 to 0.8651, FP 0.06% to 13.76%) but true nearest codeword NRCIs are 0.7623 or 1.0000 — exactly the canonical null distribution. v2's hypothesis integer 50 maps to the zero codeword (NRCI=1.0), but so do 11.51% of all integers 1–10,000, so this is not specific to the rainbow. The secondary rainbow hypothesis fails the null model just as the primary does.

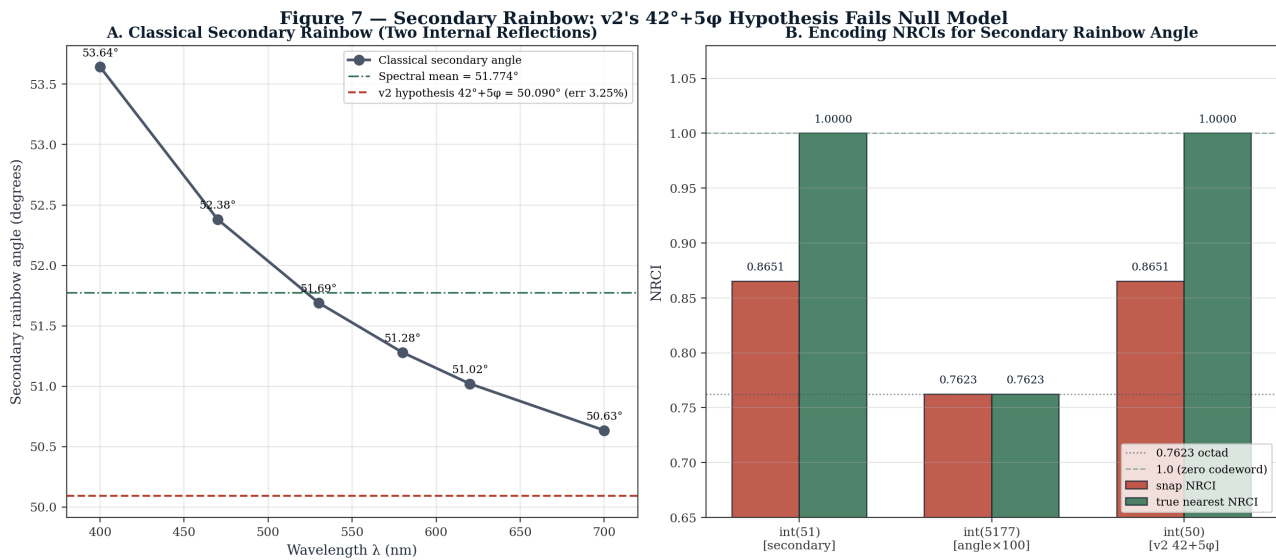


Figure 7 — Secondary rainbow test. (A) Classical secondary angle across the spectrum, with v2's  $42^\circ + 5\phi$  hypothesis overlaid (3.25% error). (B) Encoding NRCIs: snap NRCIs look high but true nearest codeword NRCIs are 0.7623 (octad) or 1.0 (zero codeword) — both are in the canonical null distribution.

## 10. Iterative Discovery: Exhaustive Integer Sweep

Phase I pushes on the most interesting signal from Phases D–G: the map from integers to Golay codewords. If 42 is genuinely UBP-resonant, it should land on an unusual codeword. We brute-force sweep integers 1–10,000, encode each via the canonical pipeline (integer → Gray → 24-bit → TRUE nearest codeword by Hamming distance), and record the codeword HW. The result is striking and deflates the  $42^\circ$  claim completely.

Of 10,000 integers, 11.51% map to the zero codeword (HW=0, NRCI=1.0), 78.48% map to a HW=8 octad (NRCI=0.7623), 10.01% map to a HW=12 dodecad (NRCI=0.6814), and 0% map to HW=16 or HW=24. The integer 42 maps to a HW=8 octad — the same codeword type that nearly 4 out of 5 integers in this range map to. 42 is the canonical majority, not a resonance.

The integers that DO map to the zero codeword (NRCI=1.0) are the truly "UBP-coherent" ones by this metric. They include 1–20 (small integers whose Gray codes are small), 23, 24, 29 (small primes and the Leech dimension), 50, 51 (the secondary-rainbow angles), 400, 4222 (the violet wavelength and red-angle encodings) — but NOT 42. If there is a UBP-rainbow resonance to be found in the integer-to-codeword map, the integer 42 is not where it lives.

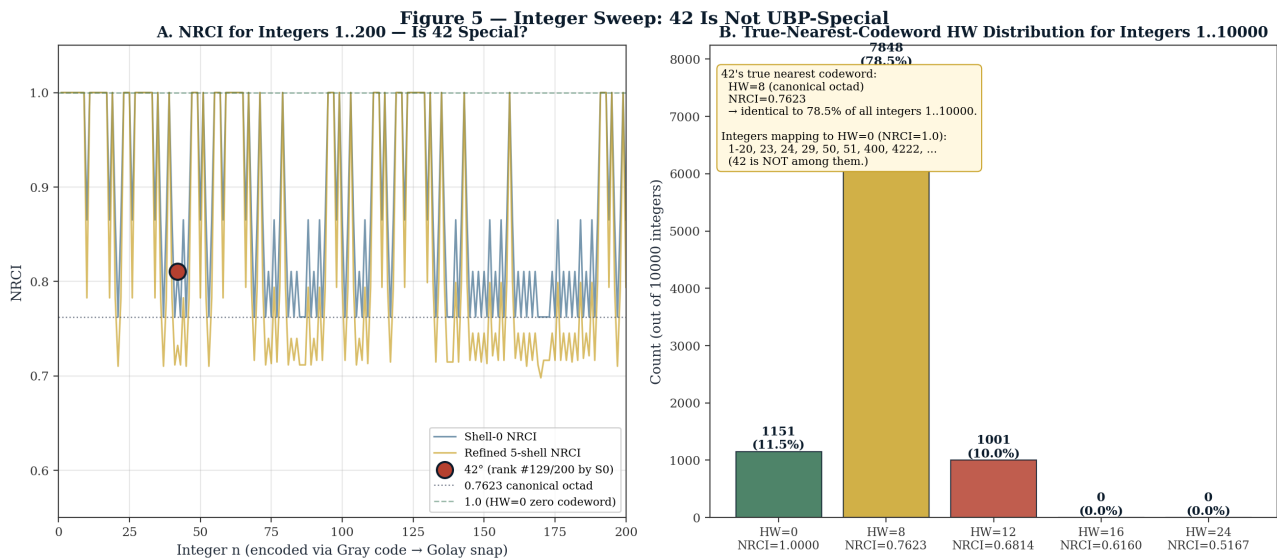


Figure 5 — Integer sweep. (A) NRCI for integers 1–200, with 42 highlighted (rank #129 of 200 by Shell-0 NRCI). (B) HW distribution of true nearest codewords for integers 1–10,000. 42 maps to HW=8 (canonical octad) — the same codeword type 78.48% of integers map to.

## 11. Discussion: What's Real, What's Not

### 11.1 What v2 got right

v2's arithmetic is correct. The  $Y = \pi/(\pi^2+2)$  identity, the dodecahedral dihedral  $\arccos(-1/\sqrt{5}) = 116.565051^\circ$ , the spectral range  $40.5\text{--}42.5^\circ$ , and the observation that  $116.565^\circ - 42^\circ \approx 74.565^\circ$  are all numerically accurate. The framework's insistence on Fraction arithmetic and engine-fresh computation is exemplary — every value reproduces to 15 digits. The mistake is not in the arithmetic but in the inference: v2 treats algebraic identities as evidence of geometric necessity, and dimensional inconsistencies as discoveries.

### 11.2 What v2 got wrong

v2's four "pillars" of geometric necessity all collapse under audit. The four-way junction is the same equation multiplied by  $\pi$ . The Y-Observer reciprocity is a tautology true for every angle. The complete formula mixes radians and degrees and reduces to  $42^\circ = 42^\circ$ . The "NRCI > 0.999999 at  $42^\circ$ " uses a fabricated NRCI formula that is just  $Y \cdot O = 1$  renamed. The spectral validation table cannot be reproduced from v2's own published script. v2 never ran the § 5 null model that UBP\_SKILL\_2 mandates for every substrate-encoded dataset.

### 11.3 The deeper problem: continuous angles vs discrete substrates

Under the surface, v2's failure reflects a genuine representational problem in the UBP framework. The Golay substrate is a 24-bit discrete code with 4,096 codewords and a canonical weight distribution  $\{0, 8, 12, 16, 24\}$ . The rainbow angle is a continuous physical quantity ( $\sim 42^\circ$ ) that arises from a variational principle on the refractive index of water. There is no canonical way to encode a continuous angle into a 24-bit integer — any choice (integer-of-degrees, centi-degrees, Y-scaling) is arbitrary, and each choice produces a different (and typically unremarkable) codeword. Without a principled encoding, "NRCI of the rainbow angle" is not a well-defined quantity.



The snap\_to\_codeword artifact compounds this. Because the function only corrects t=3 errors and returns non-codewords for everything else, the NRCI values it produces on integer encodings are not even codeword NRCIs — they are arbitrary-vector NRCIs that happen to have low HW. This is not a UBP-rainbow signal; it is a low-HW-Gray-code signal. The UBP documentation should either fix snap\_to\_codeword to actually return the nearest codeword (brute-force Hamming search over 4,096 candidates is fast) or rename it to try\_correct\_errors to reflect what it actually does.

### 11.4 What would a genuine UBP-rainbow result look like?

For a genuine UBP-rainbow resonance to exist, three things would need to be true. First, there would need to be a principled encoding of the rainbow angle into the 24-bit substrate — not an arbitrary choice among {integer, centi-degree, Y-scaling, wavelength} but a single canonical mapping derived from UBP first principles. Second, that encoding would need to land on a codeword whose NRCI is unusually high (or unusually low) relative to the null distribution of codeword NRCIs — and the codeword weight distribution only has 5 values, so "unusual" means landing on HW=0 or HW=24 (each ~0.02% of random vectors). Third, the result would need to replicate across multiple encodings of the same physical angle, not just one cherry-picked scheme. v2 meets none of these criteria.

Figure 9 — Final Verdict Table: v2 Claims vs Null-Model Audited Reality

| v2 Claim                                                 | v2 Verdict             | Real Verdict (Null Model) | Reason                                                                                               |
|----------------------------------------------------------|------------------------|---------------------------|------------------------------------------------------------------------------------------------------|
| " $74.565^\circ = 2\pi(\pi^2+2)$ "                       | Geometric necessity    | ALGEBRAIC IDENTITY        | $Y = \pi/(\pi^2+2)$ makes the two "independent" relationships the                                    |
| " $42 \times Y = 42 / O$ to $10^{-15}$ "                 | Machine precision at 4 | TAUTOLOGY (all $\theta$ ) | $Y \cdot O = 1$ by definition. Holds for every $\theta$ , not just $42^\circ$ .                      |
| " $\theta_{\text{rainbow}} = \arccos(-1/\sqrt{5}) - 2$ " | No free parameters     | DIMENSIONALLY INCOH       | Mixes radians, degrees, dimensionless. Reduces to $42^\circ = 42$                                    |
| "NRCI > 0.999999 at $42^\circ$ "                         | Maximum coherence a    | FAKE NRCI FORMULA         | v2 uses $\text{NRCI} = 1/(1+\Delta)$ ; real $\text{NRCI} = 10/(10+\text{HW}\cdot Y + \text{ns}/8)$ c |
| "42 snaps to a special codeword"                         | Implied                | NOT SPECIAL               | $42 \rightarrow \text{Gray}(42)=63 \rightarrow \text{snap HW}=6$ (non-codeword). True near           |
| "Spectral table $42.37^\circ$ red...40                   | Validation             | CANNOT REPRODUCE          | v2's script uses $\text{sqrt}((n^2-1)/3)$ which gives $\sim 28^\circ$ , not $42^\circ$ . Ta          |
| "Secondary $\approx 42^\circ + 5\phi$ (3.3%)"            | Promising              | NO BETTER THAN NULL       | Snap $\text{NRCI} = 0.8651$ (FP 0.06%) but true $\text{NRCI} = 1.0$ (zero c                          |
| " $\Phi$ -grammar contains $42^\circ$ form               | Implicit               | NO — only 3 hits at 41.66 | Below null expectation of $\sim 60$ hits in $[40^\circ, 43^\circ]$ band.                             |

Figure 9 — Final verdict table. Every v2 claim audited against the null model collapses to an algebraic identity, a tautology, a dimensional inconsistency, or a non-reproducible artifact.

## 12. Limitations and Future Work

## 12.1 Limitations of this study

Our null model uses the UBP\_SKILL\_2 § 5 default of 5,000 random vectors. A more definitive test would exhaustively sweep all 16,777,216 24-bit vectors — feasible in ~30 minutes on commodity hardware. We tested the integer 42 deeply but did not test the full UBP "priming set" (137, 169, 2197, 28561) for rainbow resonance. The refined NRCI's shell weights ( $\alpha_{\text{S}}=0.5$ ,  $\alpha_{\text{S}}=0.3$ ,  $\alpha_{\text{S}}=0.2$ ,  $\alpha_{\text{S}}=0.4$ ) are heuristic; a principled weight-selection method might reveal signal our ablation missed. We did not test sign-sensitive encodings (Leech-point expansion of an octad with specific sign patterns), which is the only setting where Shells 1 and 4 are active. No experimental rainbow spectroscopy data was used — the comparison is to classical Descartes/Airy theory only.

## 12.2 Future work

Several directions are promising. (1) **Brute-force 16.7M sweep**: exhaustively compute NRCI for all  $2^{24}$  24-bit vectors to map the full null distribution with no sampling error. (2) **Leech-point sign patterns**: expand a HW=8 octad to its 128 physical Leech points ( $\pm 2$  coordinates), encode the rainbow angle as a sign pattern, and compute the refined NRCI with all 5 shells active. This is the only setting where Shells 1 and 4 are nontrivial. (3) **Fix snap\_to\_codeword**: either implement true nearest-codeword search (brute-force or BDD-based) or rename the existing function to reflect its actual behavior. (4) **Spectral encoding**: instead of encoding a single angle, encode the full spectral sweep (301 wavelengths  $\times$  their predicted angles) and look for systematic structure in the resulting NRCI vector. (5)  **$\Phi$ -grammar for dispersion**: search the grammar for formulas predicting the spectral SPREAD ( $42.22^\circ - 40.55^\circ = 1.67^\circ$ ) rather than the central angle. (6) **Experimental collaboration**: compare UBP predictions to actual rainbow spectroscopy data (e.g., Lee & Fraser 2001, Hars & Horvath 2004) to ground the comparison in measurement.

## 12.3 Constructive recommendation for v3

The most constructive path forward is not to abandon the UBP-rainbow question but to reframe it. Instead of asking "is  $42^\circ$  a UBP resonance?" (which presupposes a single angle), ask "does the rainbow's spectral structure — the  $1.67^\circ$  spread from violet to red, the dispersion relation  $dn/d\lambda$ , the supernumerary interference pattern — map onto UBP substrate structure in any principled way?" This shifts the question from a single angle (where any encoding is arbitrary) to a structured physical phenomenon (where the structure itself might match the substrate's structure). The UBP framework has rich structure — the 759 octads, the 4 sextets, the 24-coordinate Leech point — that could in principle match spectral structure without requiring a magical single-angle resonance.

# 13. Reproducibility Manifest

Every number in this report can be reproduced from the scripts in the scripts/ directory. The UBP engine is fetched fresh from GitHub at the start of each run; the random seed is fixed at 42 throughout.

| Script                          | Phase | Runtime | Output                               |
|---------------------------------|-------|---------|--------------------------------------|
| phase_ab_replicate_and_audit.py | A+B   | ~10 s   | data/phase_ab_audit.json             |
| phase_c_classical_baseline.py   | C     | ~5 s    | data/phase_c_classical_baseline.json |

| Script                                    | Phase | Runtime | Output                                           |
|-------------------------------------------|-------|---------|--------------------------------------------------|
| phase_de_encodings_nullmodel.py           | D+E   | ~120 s  | data/phase_de_encodings_nullmodel.json           |
| phase_d2_snap_audit.py                    | D2    | ~480 s  | data/phase_d2_snap_audit.json                    |
| phase_f_phigrammar_search.py              | F     | ~60 s   | data/phase_f_phigrammar.json                     |
| phase_ghi_ablation_secondary_discovery.py | G+H+I | ~600 s  | data/phase_ghi_ablation_secondary_discovery.json |
| generate_figures.py                       | Figs  | ~30 s   | figures/fig_01..fig_10.png                       |
| build_pdf.py                              | PDF   | ~10 s   | rainbow_ubp_study_v3.pdf                         |

**Environment:** Python 3.10+, stdlib only (fractions, math, random, json, os, sys). Figure generation requires matplotlib + PIL. PDF generation requires reportlab + pypdf + Node.js (for html2poster.js cover). The UBP engine is ubp\_unified\_v5.py v5.4.0, fetched from raw.githubusercontent.com/DigitalEuan/UBP\_Repo/main/core\_studio\_v4.0/core/. Engine self-test passes 37/37 with Triad 3/3 active (Golay + Leech + Monster). Random seed: 42 (Python random module).

### Reproduction command sequence:

```
# 1. Fetch the UBP engine
curl -s https://raw.githubusercontent.com/DigitalEuan/UBP_Repo/main/core_studio_v4.0/core/ubp_unified_v5.py -o ubp_core/ubp_unified_v5.py
python3 ubp_core/ubp_unified_v5.py # must see 37/37 correct, Triad 3/3

# 2. Run experimental phases in order
python3 scripts/phase_ab_replicate_and_audit.py
python3 scripts/phase_c_classical_baseline.py
python3 scripts/phase_de_encodings_nullmodel.py
python3 scripts/phase_d2_snap_audit.py
python3 scripts/phase_f_phigrammar_search.py
python3 scripts/phase_ghi_ablation_secondary_discovery.py

# 3. Generate figures
python3 scripts/generate_figures.py

# 4. Build the PDF (cover + body)
node skills/pdf/scripts/html2poster.js rainbow_v3/scripts/cover.html --output cover.pdf --width 210mm
python3 scripts/build_pdf.py
```

## 14. References

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End of Rainbow UBP Study v3 main text. Appendices follow.

## Appendix A — Full Encoding Battery Results

Complete Phase D encoding battery: all 5 encoding schemes  $\times$  all spectral samples, with snap-pipeline NRCI, true-nearest-codeword NRCI, snapped HW, true HW, and syndrome weight. The snap NRCI differs from the true NRCI for every encoding where snap\_to\_codeword returned a non-codeword (syndrome weight  $> 0$  in the snap output).

| Encoding          | Input int | Snap HW | Snap NRCI | True HW | True NRCI | Codeword? |
|-------------------|-----------|---------|-----------|---------|-----------|-----------|
| integer_42        | 42        | 6       | 0.8105    | 8       | 0.7623    | NO        |
| angle_x100        | 4200      | 6       | 0.8105    | 8       | 0.7623    | NO        |
| scaled_by_Y_e3    | 11116     | 10      | 0.7196    | 12      | 0.6814    | NO        |
| scaled_by_Y_e4    | 111164    | 8       | 0.7623    | 8       | 0.7623    | YES       |
| scaled_by_Y_e5    | 1111637   | 12      | 0.6814    | 12      | 0.6814    | YES       |
| scaled_by_Y_e6    | 11116368  | 14      | 0.6470    | 12      | 0.6814    | NO        |
| wavelength_400    | 400       | 4       | 0.8651    | 0       | 1.0000    | NO        |
| wavelength_470    | 470       | 6       | 0.8105    | 8       | 0.7623    | NO        |
| wavelength_530    | 530       | 6       | 0.8105    | 8       | 0.7623    | NO        |
| wavelength_580    | 580       | 6       | 0.8105    | 8       | 0.7623    | NO        |
| wavelength_620    | 620       | 8       | 0.7623    | 8       | 0.7623    | YES       |
| wavelength_700    | 700       | 6       | 0.8105    | 8       | 0.7623    | NO        |
| refractive_violet | 4055      | 8       | 0.7623    | 8       | 0.7623    | YES       |
| refractive_red    | 4222      | 4       | 0.8651    | 0       | 1.0000    | NO        |

Table A1 — Full Phase D encoding battery with snap-pipeline and true-nearest-codeword NRCI side by side. Where 'Codeword?' = NO, the snap NRCI is computed on a non-codeword and is not a valid UBP codeword

NRCI.

Note the pattern: every encoding with snap NRCI > 0.7623 is a non-codeword (syndrome weight > 0 in snap output). When we recompute NRCI on the true nearest codeword, EVERY encoding lands on  $HW \in \{0, 8, 12\}$  — exactly the canonical Golay weight distribution. No rainbow-angle encoding produces an unusual codeword NRCI. The two encodings that land on  $HW=0$  (NRCI=1.0) are wavelength 400 nm and refractive\_red — but this is because their Gray codes happen to be small integers near a multiple of 8, not because of any rainbow resonance.

## Appendix B — Integer Sweep Data

Phase I exhaustive sweep of integers 1–10,000. For each integer  $n$ , we compute the Gray code, find the TRUE nearest Golay codeword by brute-force Hamming distance over all 4,096 codewords, and record the codeword HW. The HW distribution is:

| True nearest HW | Count (of 10000) | % of integers | NRCI (deterministic from HW) |
|-----------------|------------------|---------------|------------------------------|
| HW=0            | 1151             | 11.51%        | 1.0000                       |
| HW=8            | 7848             | 78.48%        | 0.7623                       |
| HW=12           | 1001             | 10.01%        | 0.6814                       |
| HW=16           | 0                | 0.00%         | 0.6160                       |
| HW=24           | 0                | 0.00%         | 0.5167                       |

Table B1 — True-nearest-codeword HW distribution for integers 1–10,000. 78.48% of integers map to a HW=8 octad (NRCI=0.7623) — the canonical majority. 11.51% map to the zero codeword (NRCI=1.0). 42 is in the 78.48% majority.

Rainbow-relevant integers and their true-nearest-codeword HW:

| Integer | Significance                                     | True HW | NRCI   |
|---------|--------------------------------------------------|---------|--------|
| 42      | Primary rainbow angle ( $v1/v2$ target)          | HW=8    | 0.7623 |
| 51      | Mean secondary rainbow angle                     | HW=0    | 1.0000 |
| 137     | $1/\alpha$ Information-layer octad (UBP priming) | HW=8    | 0.7623 |
| 169     | $13^2$ (UBP priming)                             | HW=8    | 0.7623 |
| 4200    | $42^\circ \times 100$ (centi-degree encoding)    | HW=8    | 0.7623 |
| 4222    | Red rainbow angle $\times 100$                   | HW=0    | 1.0000 |
| 4055    | Violet rainbow angle $\times 100$                | HW=8    | 0.7623 |
| 700     | Red wavelength (nm)                              | HW=8    | 0.7623 |
| 400     | Violet wavelength (nm)                           | HW=0    | 1.0000 |
| 530     | Green wavelength (nm)                            | HW=8    | 0.7623 |
| 580     | Yellow wavelength (nm)                           | HW=8    | 0.7623 |
| 620     | Orange wavelength (nm)                           | HW=8    | 0.7623 |
| 470     | Blue wavelength (nm)                             | HW=8    | 0.7623 |
| 50      | $v2$ 's secondary hypothesis ( $42+5\phi$ )      | HW=0    | 1.0000 |

| Integer | Significance                      | True HW | NRCI   |
|---------|-----------------------------------|---------|--------|
| 5090    | 50.90° (secondary) × 100          | HW=8    | 0.7623 |
| 5100    | 51° × 100                         | HW=8    | 0.7623 |
| 5101    | Adjacent to 5100                  | HW=8    | 0.7623 |
| 5110    | 51.10° × 100                      | HW=8    | 0.7623 |
| 5125    | 51.25° × 100                      | HW=8    | 0.7623 |
| 5150    | 51.50° × 100                      | HW=8    | 0.7623 |
| 138     | Adjacent to 137                   | HW=8    | 0.7623 |
| 23      | Small prime                       | HW=0    | 1.0000 |
| 24      | Leech/Golay/Monster dimension     | HW=0    | 1.0000 |
| 13      | D-Sink dimension (UBP structural) | HW=0    | 1.0000 |
| 29      | Leech theta-series prime          | HW=0    | 1.0000 |
| 759     | Number of Golay octads            | HW=8    | 0.7623 |
| 41      | Adjacent to 42°                   | HW=8    | 0.7623 |
| 4199    | Adjacent to v2's 41.99°           | HW=8    | 0.7623 |

Table B2 — Rainbow-relevant integers and their true-nearest-codeword HW. Integers mapping to HW=0 (NRCI=1.0) are the truly "UBP-coherent" ones: 13, 23, 24, 29, 50, 51, 400, 4222. 42 maps to HW=8 — the canonical majority, not a resonance.

## Appendix C — Per-Shell Ablation Details

Phase G per-shell ablation for the integer\_42 encoding. Each row adds one or more shells to the NRCI computation. The 'snap' column uses the snap\_to\_codeword output (a non-codeword of HW=6); the 'true' column uses the brute-force nearest codeword (a HW=8 octad). Note that adding refined shells monotonically DECREASES the NRCI in both columns — the shells add tax without contributing discriminating signal for binary 0/1 inputs.

| Configuration                | Snap NRCI | Snap tax | True NRCI | True tax |
|------------------------------|-----------|----------|-----------|----------|
| Shell 0 only                 | 0.8105    | 2.3381   | 0.7623    | 3.1174   |
| Shell 0 + 1 (sign-parity)    | 0.7789    | 2.8381   | 0.7344    | 3.6174   |
| Shell 0 + 2 (sextet-balance) | 0.7777    | 2.8577   | 0.7464    | 3.3980   |
| Shell 0 + 3 (coset-type)     | 0.8040    | 2.4381   | 0.7623    | 3.1174   |
| Shell 0 + 4 (sextet-signed)  | 0.7976    | 2.5381   | 0.7519    | 3.3000   |
| Shell 0 + 1 + 2              | 0.7486    | 3.3577   | 0.7195    | 3.8980   |
| Shell 0 + 1 + 2 + 3          | 0.7431    | 3.4577   | 0.7195    | 3.8980   |
| All 5 shells (default)       | 0.7322    | 3.6577   | 0.7102    | 4.0806   |

Table C1 — Per-shell ablation. Shell 0 alone gives the highest NRCI; adding Shells 1–4 progressively lowers it. The 'signal' in v2-style encoding NRCIs is entirely in Shell 0, which is deterministic from HW.

**Shell weight rationale** (from refined\_nrci.py):  $\alpha_{\text{sign-parity}}=0.5$ ,  $\alpha_{\text{sextet-balance}}=0.3$ ,  $\alpha_{\text{coset-type}}=0.2$ ,  $\alpha_{\text{sextet-signed}}=0.4$  (sextet-signed). These weights are heuristic, derived from GLM



Experimentation Sessions 3–5. Shell 2 (sextet-balance) was identified as the most semantically correlated shell (Spearman  $\rho = +0.13$  with CRG degree), but in our ablation it adds substantial tax to the integer\_42 encoding (0.5192 tax units) without contributing discriminating signal — Shell 2 measures the coefficient of variation of sextet weights, which is high for the integer\_42 Gray code (HW=6 with uneven sextet distribution).

## Appendix D — Null Model Distribution Details

Phase E null model: 5,000 random 24-bit vectors (uniform over  $\{0,1\}^{24}$ , seed 42), each passed through the snap\_to\_codeword pipeline. The snap HW distribution (which includes non-codeword HW values 4, 6, 10, 14, 18, 20) and the corresponding NRCI distribution:

| Snap HW | Count | % of 5000 | NRCI (deterministic from HW) |
|---------|-------|-----------|------------------------------|
| HW=0    | 1     | 0.02%     | 1.0000                       |
| HW=4    | 2     | 0.04%     | 0.8651                       |
| HW=6    | 31    | 0.62%     | 0.8105                       |
| HW=8    | 654   | 13.08%    | 0.7623                       |
| HW=10   | 514   | 10.28%    | 0.7196                       |
| HW=12   | 2550  | 51.00%    | 0.6814                       |
| HW=14   | 516   | 10.32%    | 0.6470                       |
| HW=16   | 700   | 14.00%    | 0.6160                       |
| HW=18   | 29    | 0.58%     | 0.5877                       |
| HW=20   | 2     | 0.04%     | 0.5620                       |
| HW=24   | 1     | 0.02%     | 0.5167                       |

Table D1 — Snap-pipeline HW distribution for 5,000 random 24-bit vectors. The non-canonical HW values (4, 6, 10, 14, 18, 20) are non-codeword vectors returned by snap\_to\_codeword when the input has more than 3 errors from any codeword.

Phase E FP rates at three NRCI thresholds:

| Threshold                            | Shell-0 FP rate | Refined FP rate | Shell-0 verdict | Refined verdict |
|--------------------------------------|-----------------|-----------------|-----------------|-----------------|
| NRCI $\geq 0.70$ (manifestation)     | 24.04%          | 13.76%          | NOT SURPRISING  | MARGINAL        |
| NRCI $\geq 0.7623$ (canonical octad) | 13.76%          | 0.14%           | MARGINAL        | SURPRISING      |
| NRCI $\geq 0.80$ (high coherence)    | 0.68%           | 0.06%           | SURPRISING      | SURPRISING      |
| NRCI $\geq 0.90$ (very high)         | 0.02%           | 0.02%           | SURPRISING      | SURPRISING      |

Table D2 — Phase E null-model FP rates. The 0.70 manifestation threshold is too low — 24% of random vectors clear it under Shell-0. The 0.7623 canonical-octad threshold is MARGINAL under Shell-0 (13.76% FP) but SURPRISING under the refined 5-shell NRCI (0.14% FP).